

Question				Marking details	Marks available				Maths	Prac
					AO1	AO2	AO3	Total		
1	(a)	(i)		Energy (or work) /charge or energy (or work) per coulomb [accept joules per coulomb] (1) The above put correctly into context, e.g. Energy given by battery / energy transfer from chemical [or to electrical]. [Free standing mark] (1)	1 1			2		
		(ii)		Method 1 $I = \frac{4.33 \text{ V}}{6.60 \Omega} [= 0.656 \text{ A}]$ or by implication (1) $r = \frac{4.80[\text{V}] - 4.33[\text{V}]}{0.656[\text{A}]} = 0.71 / 0.72 / 0.7 \Omega$ (1) Method 2 Any correct and relevant pot div equation e.g. $\frac{r}{6.6 \Omega} = \frac{4.80 \text{ V} - 4.33 \text{ V}}{4.33 \text{ V}}$ or $4.33 = \frac{4.80 \times 6.60}{r + 6.60}$ (1) $r = 0.71 / 0.72 / 0.7 \Omega$ (1) See additional guidance for methods assuming 0.7 Ω	1	1		2	2	
		(iii)	I	More current (1) Therefore greater Ir [accept: greater lost volts] (1) NB $V = E - Ir$ on its own doesn't score. or Ratio (ext res)/ r lower (1) So ratio (ext pd)/(pd across r) less (1) or equivalents	1	1		2		
			II	Current in either resistor = $\frac{3.35}{3.30} [= 1.02 / 1.015 \text{ A}]$ (1) Division by e at any stage (1) [e.g. $\rightarrow 61 \text{ C}$] Electrons in 1 minute = $3.8 \times 10^{20} \text{ C}$, ecf on I [NB electrons through parallel combination = $7.6 \times 10^{20} \rightarrow (2)$]		3		3	2	

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	(b)	(i)		$R = \frac{V^2}{P}$ or $I = \frac{1000 \text{ W}}{230 \text{ V}}$ and $R = \frac{V}{I}$ (1) $R = 53 \Omega$ [accept early rounding] (1)	1	1		2	2	
		(ii)		3.6 [MJ]		1		1	1	
	(c)			If electric heating is used rather than gas, for a given heating effect more <u>CO₂</u> produced [or more gas, a non-renewable resource [accept: fossil fuel] is used]. (1) Greater contribution to climate change/global warming [so use of electric heaters <i>should</i> be discouraged] (1) Any one of... • [But] higher cost of electricity [per kW] is discouragement enough • [But] not all power stations <i>are</i> gas-fired. [Some eg nuclear, wind, don't produce CO₂] • [But] not all homes have gas available • CO₂ absorbs long-wavelength infrared (or infrared emitted from Earth's surface) in context of climate change • Any other relevant and non-trivial point e.g. safety issue with naked flames / <u>CO</u> emission / gas leaks Example of point not worth credit: "it's none of anyone else's business how I heat my home" – the word in the question was "discouraged" / electric heaters more expensive to run			3	3		
				Question 1 total	5	7	3	15	7	0